

Honours Partial Differential Equations Notes

Edward Chernysh

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Instructor: Professor Rustum Choksi.

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Note: An alternative set of notes is available on MyCourses.¹ The only requisite for the course accompanying these notes is basic real analysis, basic undergraduate ODE theory and multivariable calculus. However, we would like to stress an intuitive understanding of the results presented in the aforementioned courses. There are two grading schemes available and these may be found on the course page. Those taking the course should note that the midterm is non-replacable.

Loosely speaking, we “analyze” a real valued function u defined on $\mathbb{R}^n$ with $n = 1, 2, 3$. A **partial differential equation** is an equation relating an unknown function u to its partial derivatives and the independent spatial variables

$$x_1, x_2, \dots, x_n.$$

The **order** of the PDE is simply the order of the *highest derivative* appearing in the equation. More often than not, the order is either one or two.

To help motivate the study of PDE we consider natural (i.e. scientific) situations in which such an equation may arise. Let $u(x_1, x_2, x_3, t)$ give the *temperature* at the spatial point (x_1, x_2, x_3) given the time t . Since heat flows from hot to cold, it is clear that there is a spatial dependence on u , and the rate at which this energy transfer occurs is related to the *spatial gradient*: ∇u . By the divergence theorem, we can obtain equations involving the Laplacian $\Delta := \nabla^2$ giving birth to the theory of harmonic functions.

DEFINITION 1. Given a PDE involving a function u and a domain² Ω , a particular function u is a solution if it satisfies the PDE at all points in Ω .

Consider $n = 2$, for simplicity. A PDE may always be written in the form

$$F(x, y, u, u_x, u_y) = 0, \quad \text{in } \Omega.$$

If the reader so desires, many examples of simple PDE may be found online. In fact, there are many lists of PDE occurring in physics, chemistry and biology. Unfortunately, the PDE that occur in nature are often non-linear equations for which there is no general theory. Consider a PDE of the form

$$L(u) = \text{“all terms involving only the independent variables and derivatives”}.$$

DEFINITION 2. Assume L is as above.

- (1) The PDE is said to be linear if L is a linear operator.
- (2) The PDE is said to be nonlinear if L is non-linear.

¹The first 14 chapters of this text will be covered in this course, and not all of it.

²An open and connected subset of $\mathbb{R}^n$

By a linear operator, we mean that for any functions $u, v : \mathbb{R}^n \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$ there holds

$$L(u + cv) = L(u) + cL(v).$$

General Solution: A function that “solves” the PDE but involves arbitrary functions (often due to a lack of initial information). For example, consider the equation

$$u_x \equiv 0, \quad \text{in } \Omega \subseteq \mathbb{R}^2$$

where $u = u(x, y)$. Clearly, any solution can depend only on y and thus, each solution is of the form $u(x, y) = g(y)$. Another example would be

$$u_{xx} \equiv 0, \quad \text{in } \mathbb{R}^2.$$

If u is a solution, then $u_x = f(y)$ in the whole space. Whence we find that

$$u(x, y) = xf(y) + g(y), \quad \text{in } \mathbb{R}^2.$$

A final example:

$$u_{xy} \equiv 0, \quad \text{in } \mathbb{R}^2.$$

Since $(u_x)_y \equiv 0$ we obtain $u_x \equiv f(x)$ for some function f . Integrating once again we obtain

$$u(x, y) = F(x) + g(y), \quad \text{where } F' \equiv f.^3$$

It is now clear that, in general, we cannot possibly hope for uniqueness of solutions.

Initial Data: To solve the issue of uniqueness we need to impose initial conditions on the PDE. This often includes information regarding the behaviour of solutions on the boundary of a domain.

First Order Equations & the Method of Characteristics. We begin this part with an example of this method.

EXAMPLE 1. $au_x + bu_y = 0$ in $\mathbb{R}^2$ with $a, b \in \mathbb{R}$. Equivalently,

$$(a, b) \cdot \nabla u \equiv 0, \quad \text{in } \mathbb{R}^2.$$

This is essentially saying “the directional derivative of u in the direction of the vector (a, b) is zero”. Geometrically, this means that any solution is constant along lines with direction (a, b) . Certainly, fix a line ℓ parallel to the vector (a, b) in $\mathbb{R}^2$. Any such line is of the form

$$bx - ay = c$$

³Of course this imposes certain regularity conditions on the solutions.

where $c \in \mathbb{R}$ determines the line uniquely. The claim is that a $C^1(\mathbb{R}^2)$ solution u is constant along any such line. To see that this is so, note that we may parametrize u solely in terms of x along ℓ since $y = \frac{b}{a}x - \frac{c}{a}$. Hence, we may write for some C^1 function f :

$$u(x, y) = f(x) \quad \text{on } \ell.$$

It then suffices to show that f' vanishing entirely on this line. To see this differentiate with respect to x to obtain by the chain rule

$$f'(x) = \frac{d}{dx} u \left(x, \frac{bx}{a} - \frac{c}{a} \right) = u_x(x, y) + \frac{b}{a} u_y(x, y) = (a, b) \cdot \nabla u(x, y) = 0$$

yielding the desired result. Now, it follows from this that u depends only on the line parallel to (a, b) given. Hence, any solution is of the form

$$u(x, y) = g(bx - ay) = g(c).$$

EXAMPLE 2. We apply this to the problem

$$\begin{cases} 4u_x - 3u_y \equiv 0, & \text{in } \mathbb{R}^n, \\ u \equiv y^3, & \text{on } x = 0. \end{cases}$$

By the previous example, we know any classical⁴ solution takes the form $u(x, y) = f(-3x - 4y)$ for some function f . Now, on the y -axis we have that

$$u(x, y) = f(-3x - 4y) = f(-4y) = y^3.$$

Taking $\xi := -4y$ we find that $f(\xi) = \frac{-\xi^3}{64}$ which implies

$$u(x, y) = f(-3x - 4y) = \frac{(3x + 4y)^3}{64}.$$

EXAMPLE 3 (Time Dependent Transport Equation). By way of an example consider the problem

$$\begin{cases} u_t + cu_x \equiv 0, & \text{for } (x, t) \in \mathbb{R} \times [0, \infty), \\ u(x, 0) = g(x). \end{cases}$$

By now we know that we should rewrite this as $(c, 1) \cdot \nabla u(x, t) \equiv 0$ and that the general solution will be of the form $u(x, t) = f(x - ct)$ for some function f . Taking $t = 0$ we find that

$$u(x, 0) = f(x) = g(x)$$

it follows that $f(x) = g(x)$ and hence that $u(x, t) = g(x - ct)$.

⁴Given a PDE let α denote the order of the highest derivative. A solution u is called classical if it belongs to $C^\alpha(\mathbb{R}^n)$.

It is now natural to try and extend our knowledge to equations with variable coefficients.